Ogbonna Arinze

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SUMMARY

Senior Frontend Engineer with 8 years of professional experience building high-impact, user-centric web applications at scale. Specialized in designing and implementing intuitive, responsive interfaces using React, TypeScript, and modern state management, with hands-on experience building Chrome extensions and AI-powered products serving millions of users. Proven track record of owning features end-to-end, optimizing frontend performance, establishing testing and CI/CD best practices, and mentoring junior developers. Passionate about developer experience, product quality, and shipping fast in high-ownership startup environments.

CORE COMPETENCIES

- **Frontend & UI:** React, TypeScript, Next.js, Vue.js, Redux, Zustand, React Query, Chrome Extension APIs, responsive design, component architecture, and design systems (Radix UI, Tailwind CSS).
- **Performance & Optimization:** Lazy loading, code splitting, memoization, CDN strategies, bundle analysis (Webpack, Rollup), Lighthouse, Chrome DevTools, React Profiler, and Core Web Vitals.
- **Testing & Quality:** Playwright, Jest, Enzyme, Cypress, Storybook, Allure Reports, ESLint, Prettier, Husky, and CI-driven QA.
- **CI/CD & Infrastructure:** GitHub Actions, Docker, AWS (CloudFront, S3), Vercel, Git workflows, and automated deployment pipelines.
- **Backend & API Integration:** NodeJS, NestJS, REST APIs, GraphQL, JWT authentication, role-based access control, PostgreSQL, and MongoDB.
- **AI & LLM Systems:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, vector search (Chroma, pgvector), and agentic workflows.
- **Leadership:** Technical mentorship, architecture reviews, Agile/Scrum, peer code reviews, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer (React / TypeScript) *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing user adoption across healthcare teams.
- Maintained and extended the existing **Vue.js** codebase during the migration period, fixing bugs and shipping features in parallel with the React rebuild to ensure zero disruption to end users.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Optimized application performance by implementing **lazy loading, code splitting**, image optimization, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools**. Applied memoization and responsive layouts to minimize re-renders across devices.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript**, using HTTP-only cookies for secure token handling, protected client-side routes, and permission-based UI rendering.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages, and configuring Slack notifications for build quality visibility.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to query their data in real time via a conversational interface.
- Mentored a team of two junior developers (interns), conducting code reviews and fostering a culture of technical excellence.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and more confident releases.

- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with clear communication of component APIs.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** UI for user roles and permissions management.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components, ensuring a consistent and intuitive user interface across the application.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: NestJS API backend, **Next.js** citizen-facing web app with **React 19** and **22 interactive chart types** (Recharts), admin dashboard, document ingestion pipeline, marketing landing page, and a Remotion video generation service.
- Built the **Next.js** frontend with **Tailwind CSS v4**, dark mode, URL state management with nuqs, responsive mobile layouts, and **Playwright** E2E tests across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5).
- Designed and implemented a **multi-agent AI system** using the Mastra framework with 5 specialized agents and a hybrid **RAG pipeline** combining pgvector + BM25 full-text search with Reciprocal Rank Fusion and Cohere reranking.
- Configured **CI/CD** with **GitHub Actions** and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*